MEMORANDUM

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TO: Joseph Pinheiro, Professor, Mechanical Engineering

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SUBJECT: **Lab 8 – Flow Through a converging Diverging Nozzle**

Abstract

This experiment investigated compressible airflow through a converging-diverging nozzle connected to a four-stage centrifugal compressor. Pressure measurements were collected at the inlet, throat, and outlet of the nozzle while the compressor speed gradually increased. The measured vacuum pressures were converted to absolute pressures and used to calculate pressure ratios, mass flow rate, and experimental throat Mach number. These results were compared with incompressible and compressible one-dimensional flow models to evaluate the effect of compressibility as the flow speed increased. The experimental data showed that the throat pressure ratio decreased as mass flow rate increased, while the throat Mach number approached one. This indicated that the flow reached choked conditions near $\dot{m} \approx 0.0170\ kg/s$, with an experimental throat pressure ratio of approximately $\frac{p_{th}}{p_0} = 0.537$. This value was close to the theoretical choked pressure ratio of $\frac{p_{th}}{p_0} = 0.528$ for air. While the calculated Mach number exceeded unity at the highest flow conditions, this result is not physically consistent with true throat conditions and is likely due to the pressure tap being located slightly downstream of the actual minimum-area section of the nozzle. Consequently, the measured pressure was lower than the true throat pressure, resulting in pressure ratios below the theoretical choking limit and artificially elevated Mach number calculations. Overall, the compressible flow model provided better agreement with the experimental data at higher Mach numbers, while the incompressible model became less accurate as compressibility effects became significant.

Background/Theory

This experiment studied compressible flow through a converging-diverging nozzle. As air moves through the converging section, the decrease in area causes the velocity to increase and the static pressure to decrease. The main parameter used to describe this behavior is Mach number *M*, which is the ratio of the local flow velocity *V* to the local speed of sound *c*:

$$M = \frac{V}{c} \tag{1}$$

At low speeds, where (Ma < 0.3), density changes are small and the flow can be treated as incompressible. As the Mach number increases, compressibility effects become more important,

particularly near the throat where the flow reaches sonic conditions and choking may occur. Choked flow occurs when the Mach number at the throat reaches one. At this point, the throat flow is sonic, and the mass flow rate cannot increase further even if the outlet pressure continues to decrease. For air with (k=1.4), the theoretical throat pressure ratio at the choking condition is approximately 0.528. This value was used as a reference when comparing the observed choked condition to the compressible flow model.

The experimental mass flow rate was calculated using the pressure drop from the room to the inlet of the nozzle. This section was modeled as incompressible because the inlet velocity is much lower than the throat velocity. The resulting mass flow rate was then used with the measured throat pressure to determine the experimental throat Mach number.

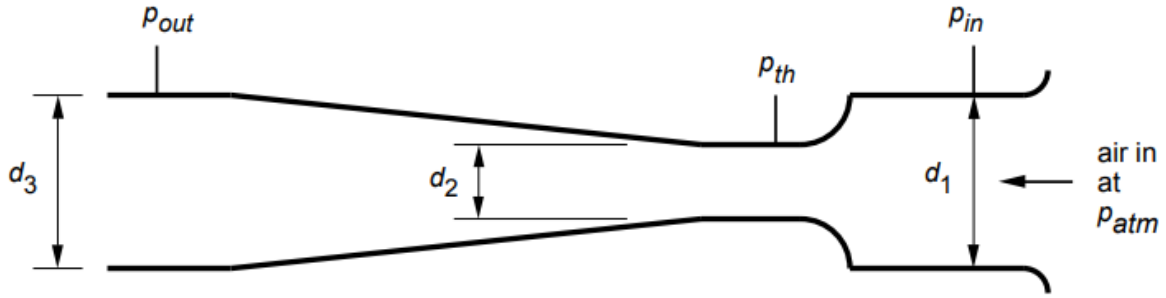


Fig 1.1 Converging-diverging nozzle picture and schematic

The data was compared to both incompressible and compressible one-dimensional flow models. The incompressible model uses Bernoulli's equation and assumes constant density, so it is expected to work best only at lower Mach numbers. The compressible model uses isentropic flow relations and accounts for changes in air density as pressure changes. Because of this, the compressible model should better match the data as the flow speed increases.

Apparatus and Experimental Procedure

The experiment was performed using the compressible flow apparatus with a converging-diverging nozzle connected to a four-stage centrifugal compressor. The nozzle had pressure taps at the inlet, throat, and outlet, which were connected to the display panel pressure transducers. The nozzle diameters were measured using a bore gage and micrometer prior to testing. Atmospheric pressure and room temperature were also recorded so that air density and pressure ratios could be calculated. The experiment was conducted in accordance with the laboratory manual.

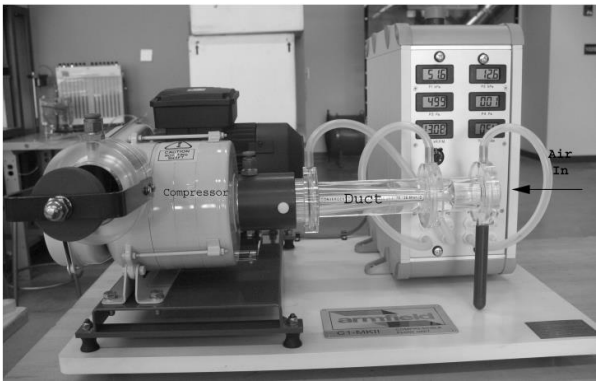


Fig 1.2 Compressible flow apparatus

The pressure readings were measured as vacuum pressures relative to atmospheric pressure, so they were later converted to absolute pressures before calculating pressure ratios.

The compressor speed was increased gradually using the speed control knob. Data collection began when the outlet pressure reading reached 0.5 kPa vacuum. The outlet pressure then increased in increments of 0.5 kPa up to 12.5 kPa. At each speed setting, the inlet, throat, and outlet pressure readings were recorded once the values became steady. During testing, observations were also made about the airflow noise through the nozzle since a change in sound was used as an indication that the flow had become choked.

After the final pressure reading was taken, the compressor speed was slowly reduced back to zero. The recorded pressure data was then used to calculate absolute pressures, pressure ratios, mass flow rate, and throat Mach number. These experimental values were compared with the incompressible and compressible flow models.

Analysis

Due to the lab apparatus measured vacuum pressure rather than absolute values, the pressure readings from the transducer could not be plugged directly into the pressure ratio equations. Since each transducer measured the pressure drop below atmospheric, the pressure readings were first converted to absolute pressures by subtracting the vacuum reading from the ambient atmospheric pressure:

$$P_{abs} = p_0 - p_{vac} \tag{2}$$

Where p_o is the atmospheric pressure and p_{vac} is the vacuum reading at the inlet, throat, or outlet. From there, calculating the pressure ratio for each location was straightforward.

$$\frac{p}{p_0} = \frac{p_{abs}}{p_0} \tag{3}$$

The mass flow rate was calculated by taking the pressure drop between the room and the nozzle inlet. Because the air moves significantly slower at the inlet than at the narrow throat, in this section it can be safely treated as incompressible flow. By factoring in the inlet area (A_{in}) and accounting for a minor contraction loss coefficient ($K_L = 0.08$) the mass flow rate is determined by:

$$\dot{m} = A_{in}\sqrt{\frac{2\rho_0(p_0 - p_{in})}{1 + K_L}} \tag{4}$$

Here, ρ_0 represents the air density at ambient room conditions, and p_{in} is the absolute inlet pressure. Once the mass flow rate was obtained, the throat Mach number (Ma_{th}) could be solved for. This calculation ties together the physical throat area (A_{th}), ambient conditions, the specific heat ratio for air ($k = 1.4$), and the empirical throat pressure ratio ($\frac{p_{th}}{p_0}$):

$$Ma_{th} = \frac{\dot{m}}{A_{th}\sqrt{k\rho_0 p_0\left(\frac{p_{th}}{p_0}\right)^{\frac{k+1}{k}}}} \tag{5}$$

Finally, these calculated values were used to evaluate how well our real-world data matched up against standard theoretical behaviors specifically comparing it to the incompressible model (which assumes air density never changes) and the compressible model (which uses isentropic flow relationships to account for density shifts).

Results

Figures 2-4 were created using the experimental data and model calculations seen in Attachments 1-3. These attachments contain pressure ratios, mass flow rates, and Mach numbers used to compare the experimental results with the incompressible and compressible flow models.

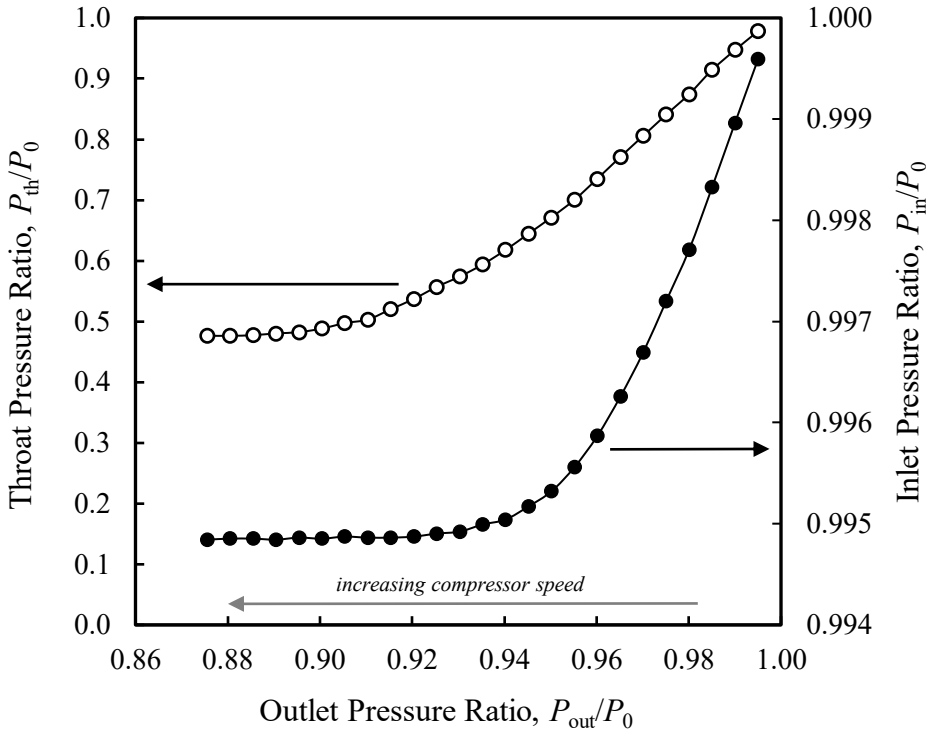


Figure 2. Pressure Ratios vs. Outlet Pressure Ratio

Figure 2 compares the throat pressure ratio and inlet pressure ratio to the outlet pressure ratio. We see that the outlet pressure decreased with an increase in compressor speed. The throat pressure ratio decreased significantly while the inlet pressure ratio was close to 1. This agrees with the 1-D compressible flow theory because as the flow accelerates towards the throat, we see the largest static pressure drop at the location of smallest area. The section of almost constant inlet pressure suggests that the inlet remained close to atmospheric conditions, while most of the pressure change occurred through the nozzle. By looking at the difference between the inlet and throat pressure we can infer that there is a presence of viscous and non-ideal flow effects. The low throat pressure ratios suggest that the flow was approaching choked conditions, this can also be seen in Figures 3 and 4 where the mass flow levels off near $\dot{m} \approx 0.017\ kg/sec$ and the Mach number approaches 1.

It should also be noted that the measured throat pressure ratio eventually decreases below the theoretical choking value of $p_{th}/p_0 = 0.528$. Since this pressure ratio corresponds to sonic conditions at the true throat, values below this limit are not physically consistent with choked throat. This observation suggests

that the pressure tap identified as the throat tap may be located slightly downstream of the actual minimum-area section, causing pressures lower than the true throat pressure to be measured. This effect is discussed further in Figures 3 and 4.

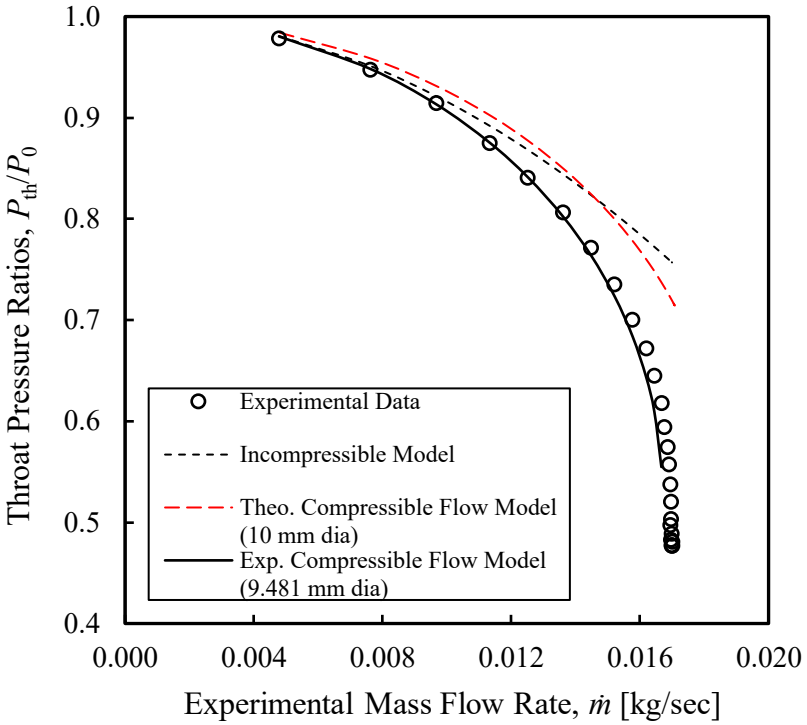


Figure 3. Throat Pressure Ratio vs. Mass Flow Rate

In Figure 3 we see that as the experimental mass flow rate increases, the throat pressure ratio decreased. This trend makes sense because we know that increasing the mass flow rate causes the flow to accelerate through the convergence of the nozzle, which in turn lowers the static pressure at the throat. Our experimental data shows that the flow approached choked conditions at about $\dot{m} = 0.0170\ kg/sec$, which corresponds to a throat pressure ratio of about 0.537. This value is similar to the inviscid 1-D compressible flow theory's value of $p_{th}/p_0 = 0.528$, which corresponds to a Mach number of 1. Additionally, a noticeable change in airflow noise was observed near this operating condition, providing further experimental evidence that choking had occurred. The compressible flow model using the throat diameter follows the experimental data closer than the incompressible model as we approach choke flow. It is important to note that the difference between the compressible flow models is due to the variation between the nominal and measured throat diameters.

While the experimental data initially follows the expected trend toward the theoretical choked-flow pressure ratio of $p_{th}/p_0 = 0.528$, the measured pressure ratio continues to decrease below this value for the highest flow conditions. For a converging-diverging nozzle, a pressure ratio of 0.528 corresponds to sonic conditions, Ma = 1, at the true throat. Once choking occurs, the pressure at the throat should not continue decreasing below this theoretical limit. However, the experimental data reaches pressure ratios as low as $p_{th}/p_0 = 0.476$. The most likely explanation for this discrepancy is that the pressure tap identified as the throat tap is located slightly downstream of the actual minimum-area section of the nozzle. If the pressure measurement is taken within the diverging section, the flow may continue to

expand and experience a further reduction in static pressure, resulting in pressure ratios below the theoretical choking value. Therefore, the pressure ratios below 0.528 are likely due to the measurement location rather than the actual conditions at the true throat.

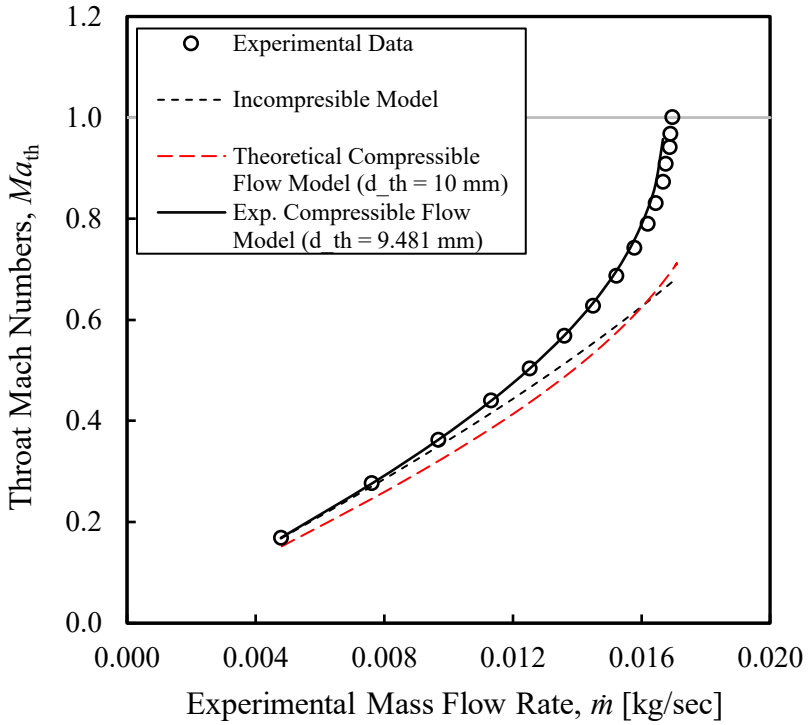


Figure 4. Throat Mach Number vs. Mass Flow Rate

Figure 4 shows that the experimental throat Mach number increased as the experimental mass flow rate increased. This trend is expected because increasing the mass flow rate requires an increase in the velocity of the air passing through the nozzle throat. Since Mach number is defined as the ratio of flow velocity to the local speed of sound, an increase in velocity results in a corresponding increase in Mach number. The experimental data approaches $Ma_{th} = 1$ near the same mass flow rate identified in Figure 3, further supporting that the flow reached choked conditions. The compressible and incompressible models begin to differ by approximately 3–5% near $Ma_{th,comp} = .44$. At this point, $Ma_{th,inc} \approx .42$, this corresponds to a difference of about 4.5%. This divergence is expected because compressibility effects become increasingly important as Mach number increases. At higher flow speeds, changes in density can no longer be neglected, causing the incompressible flow assumption to become less accurate.

As discussed in Figure 3, the measured throat pressure ratio decreases below the theoretical choking value of $p_{th}/p_0 = 0.528$, which is not physically consistent with conditions at the true throat. Because the experimental Mach number is calculated directly from the measured throat pressure ratio, these lower-than-expected pressure measurements result in calculated Mach numbers that exceed 1.0. Under choked conditions, the Mach number at the true throat should reach a maximum value of approximately 1.0. However, the experimental data predicts Mach numbers greater than one at the highest mass flow rates. The most reasonable explanation is that the pressure tap identified as the throat tap is slightly offset downstream of the actual minimum-area location. As a result, pressure measurement is taken in the

diverging section of the nozzle rather than at the true throat. Because the static pressure in this region is lower than the actual throat pressure, the calculated Mach number is artificially increased, resulting in values greater than unity. Therefore, the Mach numbers exceeding 1.0 should not be interpreted as actual supersonic flow occurring at the throat, but rather as a consequence of the pressure tap location. Despite this measurement limitation, the data still demonstrates the approach to choking, as evidenced by the throat pressure ratio approaching the theoretical choking value and the mass flow rate leveling off near 0.0170 kg/s.

Conclusion

This experiment demonstrated compressible airflow through a converging-diverging nozzle and showed that the throat pressure ratio decreased as mass flow rate and throat Mach number increased. The flow reached or closely approached choked conditions at approximately $\dot{m}$ = 0.0170 kg/s, with an experimental throat pressure ratio of approximately $p_{th}/p_0 = 0.537$, which was close to the theoretical choked-flow value of 0.528 predicted by inviscid one-dimensional compressible flow theory. Additionally, the experimental mass flow rate approached a limiting value near 0.0170 kg/s, providing further evidence that choking had occurred.

The compressible flow model agreed more closely with the experimental data than the incompressible model, especially at higher Mach numbers where density changes became significant. The divergence between the compressible and incompressible models near Math ≈ 0.44–0.45 was consistent with the expectation that compressibility effects become increasingly important as Mach number increases.

Although the calculated experimental Mach number exceeded unity at the highest flow conditions, the pressure ratio indicates that the source of the discrepancy originates with the measured throat pressure. The experimental throat pressure ratio decreased below the theoretical choking value of $p_{th}/p_0 = 0.528$, reaching values as low as 0.476. Since a pressure ratio of 0.528 corresponds to sonic conditions at the true throat, pressures below this value are not physically consistent with a choked throat. The most likely explanation is that the pressure tap used to measure the throat pressure was positioned slightly downstream of the actual minimum-area section of the nozzle. As a result, the measured pressure would be lower than the true throat pressure, leading to both artificially low-pressure ratios and artificially elevated Mach number calculations. Therefore, the apparent pressure ratios below the choking limit and Mach numbers greater than one are most likely the result of the pressure measurement location rather than actual flow conditions at the throat. Despite this limitation, the experimental data successfully demonstrated the approach to choking and showed good agreement with the theoretical behavior predicted by compressible flow theory.

References

[1] ME 347 Lab Manual

Attachments

1. Raw Data
2. Experimental Pressure Ratios at the Inlet, Throat, and Outlet
3. Throat Pressure Ratios
4. Mach Numbers vs Mass Flow Rate at Throat

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Attachment 1: Raw Data

Data No. [-]	Raw Data from Gages		
	P_{th} [kPa-v]	P_{out} [kPa-v]	P_{in} [Pa-v]
1	2.2	0.5	41
2	5.3	1.0	104
3	8.6	1.5	168
4	12.6	2.0	230
5	16.0	2.5	281
6	19.5	3.0	332
7	23.0	3.5	376
8	26.6	4.0	415
9	30.1	4.5	446
10	33.0	5.0	470
11	35.7	5.5	485
12	38.4	6.0	498
13	40.8	6.5	503
14	42.8	7.0	510
15	44.5	7.5	512
16	46.5	8.0	515
17	48.2	8.5	516
18	49.9	9.0	516
19	50.5	9.5	515
20	51.4	10.0	517
21	52.0	10.5	516
22	52.2	11.0	518
23	52.5	11.5	517
24	52.6	12.0	517
25	52.6	12.5	518

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Attachment 2 – Experimental Pressure Ratios at the Inlet, Throat, and Outlet

Data No.	$(P_{th}/P_{atm})_e$	$(P_{out}/P_{atm})_e$	$(P_{in}/P_{atm})_e$
[-]	[-]	[-]	[-]
1	0.978	0.995	1.000
2	0.947	0.990	0.999
3	0.914	0.985	0.998
4	0.875	0.980	0.998
5	0.841	0.975	0.997
6	0.806	0.970	0.997
7	0.771	0.965	0.996
8	0.735	0.960	0.996
9	0.700	0.955	0.996
10	0.671	0.950	0.995
11	0.645	0.945	0.995
12	0.618	0.940	0.995
13	0.594	0.935	0.995
14	0.574	0.930	0.995
15	0.557	0.925	0.995
16	0.537	0.920	0.995
17	0.520	0.915	0.995
18	0.503	0.910	0.995
19	0.497	0.905	0.995
20	0.488	0.900	0.995
21	0.482	0.895	0.995
22	0.480	0.890	0.995
23	0.477	0.886	0.995
24	0.476	0.881	0.995
25	0.476	0.876	0.995

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Attachment 3- Throat Pressure Ratios

Data No.	Experimental	Incompressible	Compressible (9.481mm dia)	Compressible (10 mm dia)	$\dot{m}_{exp}$
1	0.978	0.981	0.980	0.984	0.0048
2	0.947	0.951	0.948	0.958	0.0076
3	0.914	0.921	0.913	0.930	0.0097
4	0.875	0.892	0.875	0.901	0.0113
5	0.841	0.868	0.841	0.876	0.0125
6	0.806	0.844	0.802	0.848	0.0136
7	0.771	0.824	0.764	0.822	0.0145
8	0.735	0.805	0.723	0.796	0.0152
9	0.700	0.791	0.684	0.774	0.0158
10	0.671	0.779	0.644	0.756	0.0162
11	0.645	0.772	0.609	0.743	0.0164
12	0.618	0.766	0.555	0.732	0.0167
13	0.594	0.764	#N/A	0.728	0.0168
14	0.574	0.761	#N/A	0.721	0.0169
15	0.557	0.760	#N/A	0.719	0.0169
16	0.537	0.758	#N/A	0.716	0.0170
17	0.520	0.758	#N/A	0.715	0.0170
18	0.503	0.758	#N/A	0.715	0.0170
19	0.497	0.758	#N/A	0.716	0.0170
20	0.488	0.757	#N/A	0.714	0.0170
21	0.482	0.758	#N/A	0.715	0.0170
22	0.480	0.757	#N/A	0.713	0.0170
23	0.477	0.757	#N/A	0.714	0.0170
24	0.476	0.757	#N/A	0.714	0.0170
25	0.476	0.757	#N/A	0.713	0.0170

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Attachment 4 – Mach Numbers vs Mass Flow Rate at Throat

Data No.	Experimental Ma	Incompressible Ma	Compressible 9.481mm dia Ma	Compressible 10 mm dia Ma	$\dot{m}_{exp}$
1	0.16897	0.17	0.17	0.15	0.0048
2	0.27661	0.27	0.28	0.25	0.0076
3	0.36237	0.35	0.36	0.32	0.0097
4	0.44049	0.42	0.44	0.39	0.0113
5	0.50364	0.47	0.50	0.44	0.0125
6	0.56766	0.51	0.57	0.49	0.0136
7	0.62744	0.55	0.63	0.54	0.0145
8	0.68663	0.59	0.70	0.58	0.0152
9	0.74206	0.61	0.76	0.62	0.0158
10	0.78976	0.64	0.82	0.65	0.0162
11	0.83085	0.65	0.87	0.66	0.0164
12	0.87323	0.66	0.96	0.68	0.0167
13	0.90778	0.66	#N/A	0.69	0.0168
14	0.94120	0.67	#N/A	0.70	0.0169
15	0.96755	0.67	#N/A	0.70	0.0169
16	1.00114	0.67	#N/A	0.71	0.0170
17	1.03000	0.68	#N/A	0.71	0.0170
18	1.05962	0.68	#N/A	0.71	0.0170
19	1.06949	0.67	#N/A	0.71	0.0170
20	1.08839	0.68	#N/A	0.71	0.0170
21	1.09887	0.68	#N/A	0.71	0.0170
22	1.10491	0.68	#N/A	0.71	0.0170
23	1.10976	0.68	#N/A	0.71	0.0170
24	1.11175	0.68	#N/A	0.71	0.0170
25	1.11283	0.68	#N/A	0.71	0.0170